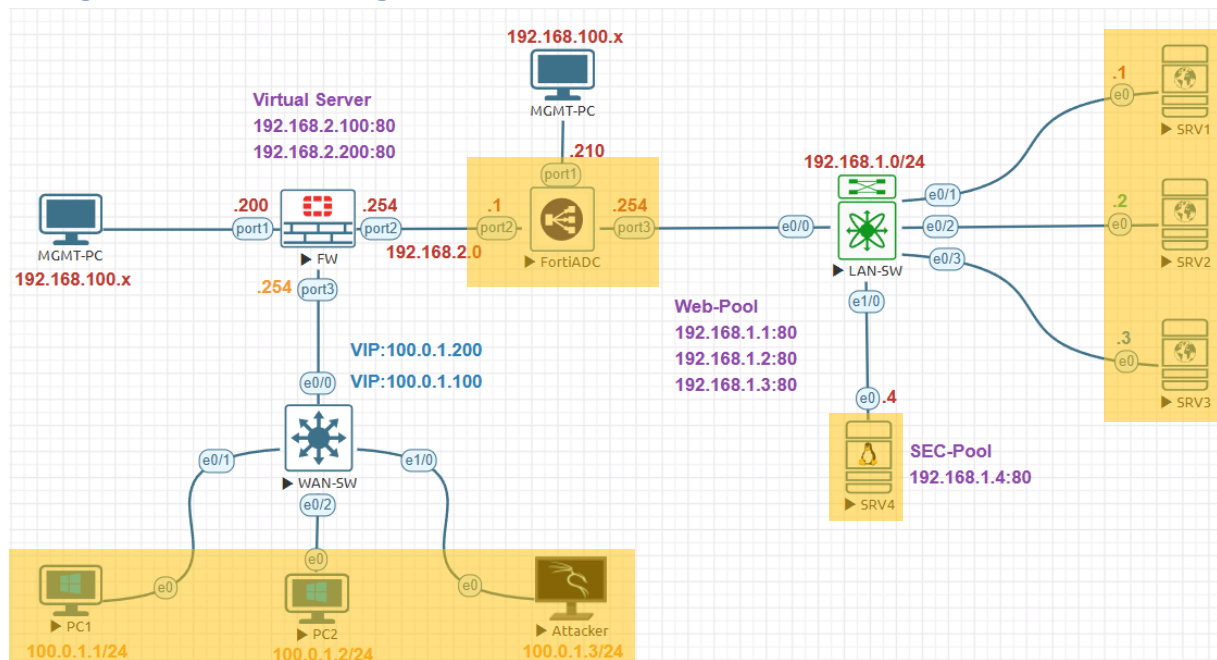
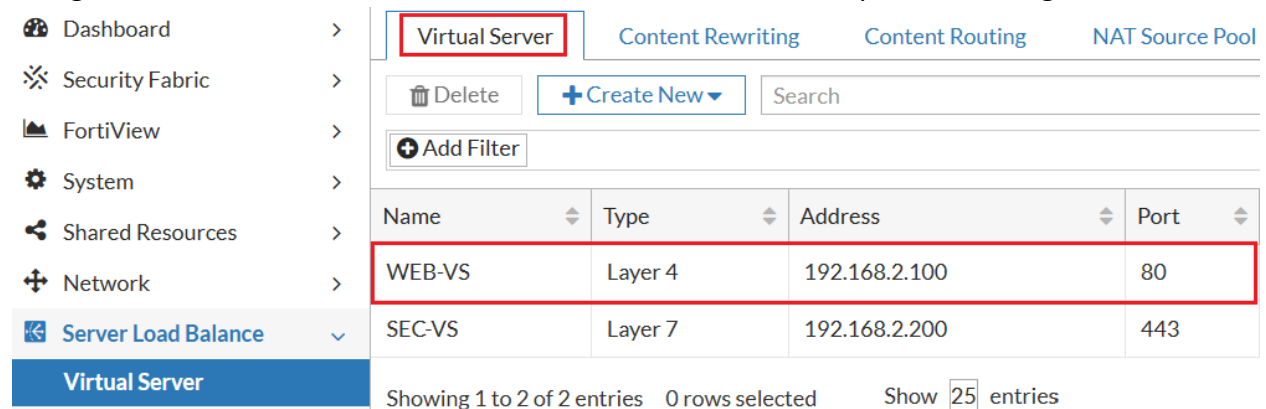


Weight Load Balancing Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Navigate to **Server Load Balance > Virtual Server** double click to open the setting.

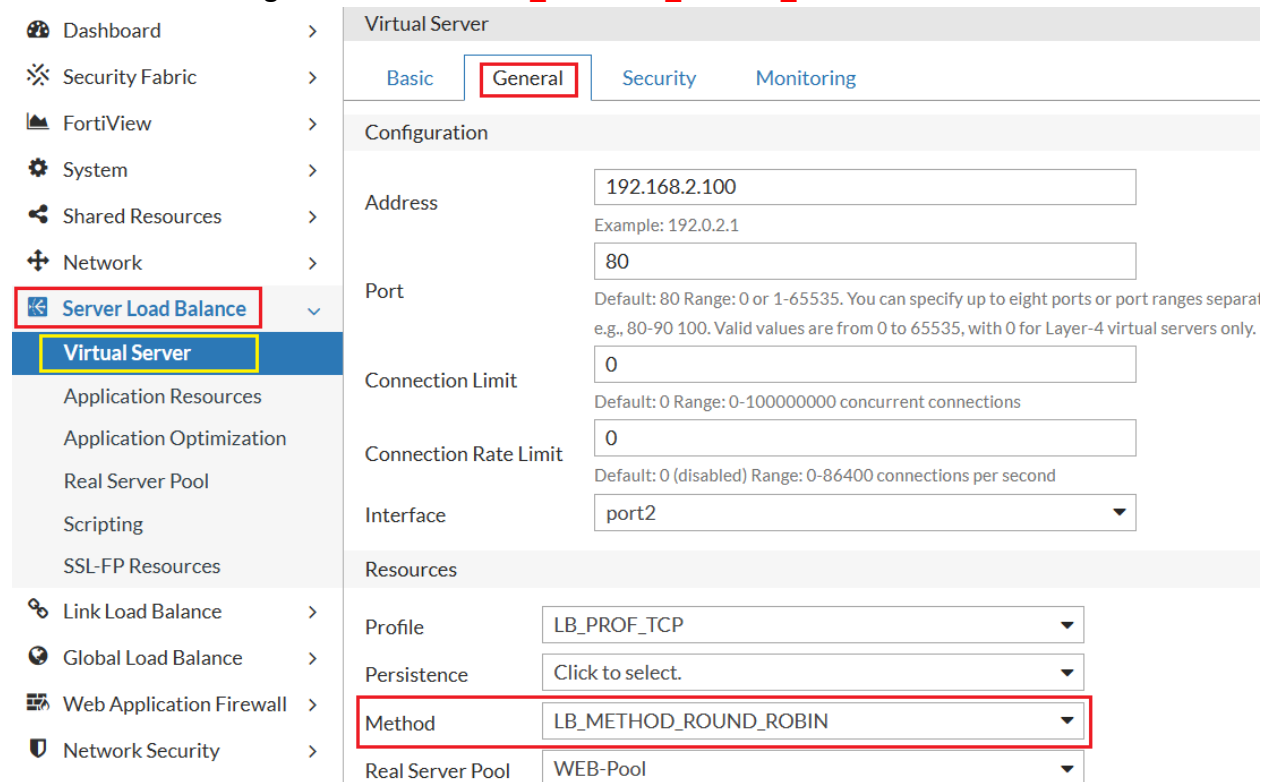


Virtual Server

Name	Type	Address	Port
WEB-VS	Layer 4	192.168.2.100	80
SEC-VS	Layer 7	192.168.2.200	443

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

Go to **General** change the Method to **LB_METHOD_ROUND_ROBIN** click **OK**.



Virtual Server

Basic General Security Monitoring

Configuration

Address 192.168.2.100
Example: 192.0.2.1

Port 80
Default: 80 Range: 0 or 1-65535. You can specify up to eight ports or port ranges separat e.g., 80-90 100. Valid values are from 0 to 65535, with 0 for Layer-4 virtual servers only.

Connection Limit 0
Default: 0 Range: 0-100000000 concurrent connections

Connection Rate Limit 0
Default: 0 (disabled) Range: 0-86400 connections per second

Interface port2

Resources

Profile LB_PROF_TCP

Persistence Click to select.

Method LB_METHOD_ROUND_ROBIN

Real Server Pool WEB-Pool

Next navigate to **Server Load Balance > Real Server Pool > Real Server Pool** tab double click on **WEB-Pool** to open the setting.

Real Server Pool Real Server Server SSL

Delete + Create New + Add Filter

Name	Address Type	Health Check
WEB-Pool	IPv4	Enable
SEC-Pool	IPv4	Disable

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

Virtual Server
Application Resources
Application Optimization
Real Server Pool

Click the **pencil** icon to edit each server one by one to change the **Weight**.

Member

Delete + Create New + Add Filter

ID	Name	Address	Health Check	Port	
1	SRV1	192.168.1.1	inherited	80	
2	SRV2	192.168.1.2	inherited	80	
3	SRV3	192.168.1.3	inherited	80	

Showing 1 to 3 of 3 entries 0 rows selected

Let's adjust the weight settings in the Real Server Pool. **SRV1** is assigned a weight of **1**, **SRV2** a weight of **2** and **SRV3** a weight of **3**. The connection distribution will follow a **1:2:3 ratios**. This means SRV3 will handle more connections compared to SRV1 and SRV2.

If you have a total of **60 connections** and you want to distribute them according to the weights. SRV1 will handle **10 connections**, SRV2 will handle **20 connections**, and SRV3 will handle **30 connections**.

Change the Weight to **1** and click **Save** to apply the setting.

Name	Port	80 Default: 80 Range: 0-65535
Address Type	Weight	1 Default: 1 Range: 1-256
Type	Recover	0 Default: 0 (disabled) Range: 0-86400 seconds
Health Check	Warm Up	0 Default: 0 (disabled) Range: 0-86400 seconds
Health Check Relative	Warm Rate	100 Default: 100 Range: 1-86400 connections per second
Health Check List	Connection Limit	0 Default: 0 (disabled) Range: 0-1048576 concurrent connections
Direct Route Mode	Connection Rate Limit	0 Default: 0 (disabled) Range: 0-86400 connections per second
Real Server SSL Profile		
		Save Cancel

Change the Weight to **2** and click **Save** to apply the setting.

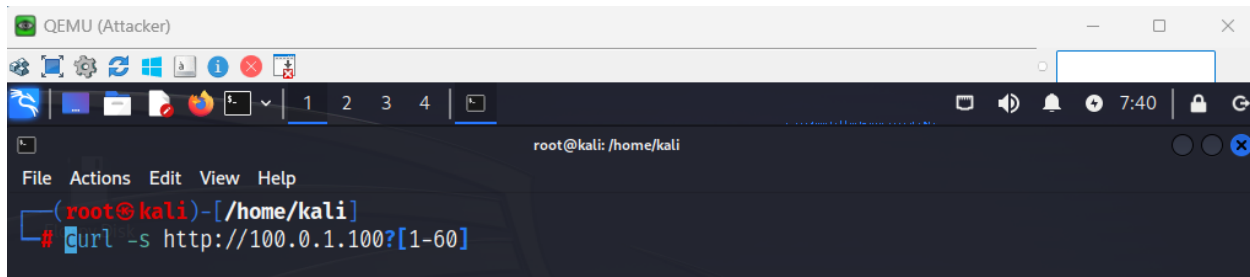
Address Type	Member	
Type	Status	<input checked="" type="button" value="Enable"/> Disable Maintain
Health Check	Real Server	SRV2 Default: 80 Range: 0-65535
Health Check Relative	Port	80 Default: 80 Range: 0-65535
Health Check List	Weight	2 Default: 1 Range: 1-256
	Recover	0 Default: 0 (disabled) Range: 0-86400 seconds

Change the Weight to **1** and click **Save** to apply the setting.

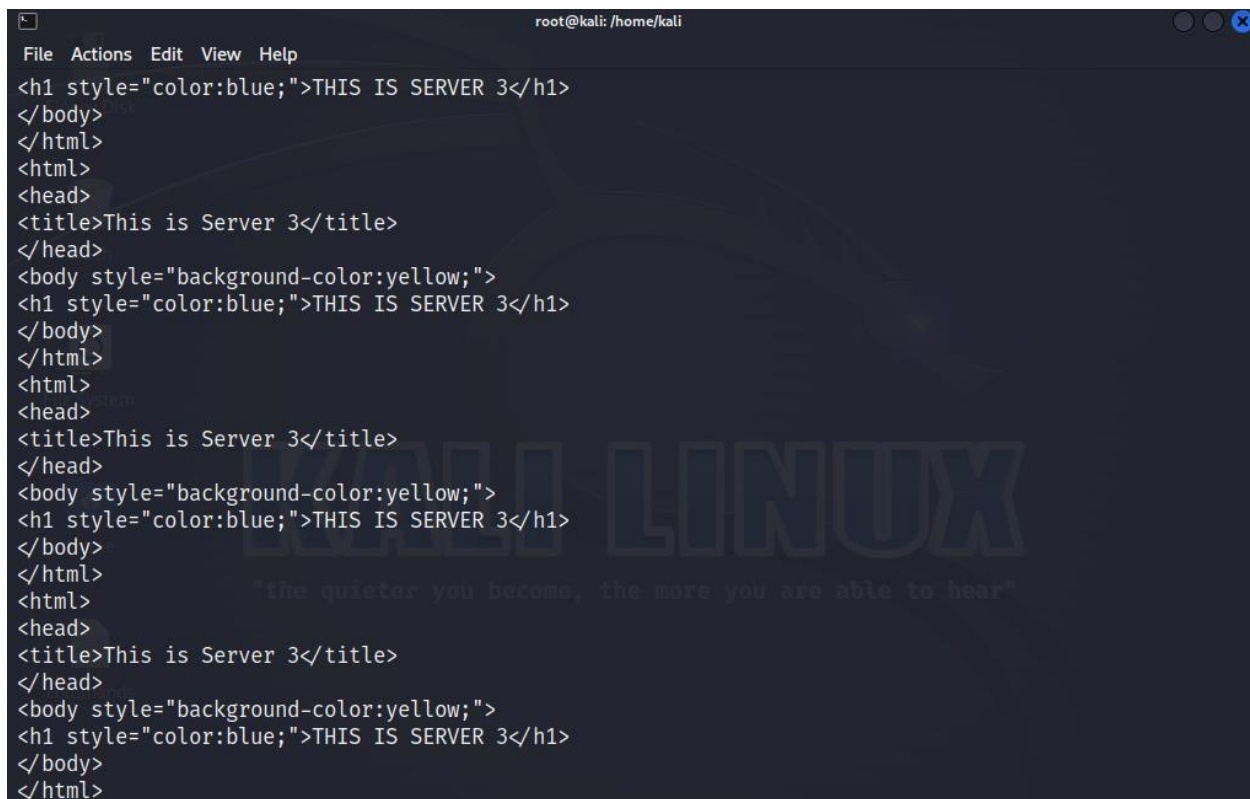
Address Type	Member	
Type	Status	<input checked="" type="button" value="Enable"/> Disable Maintain
Health Check	Real Server	SRV3 Default: 80 Range: 0-65535
Health Check Relative	Port	80 Default: 80 Range: 0-65535
Health Check List	Weight	3 Default: 1 Range: 1-256
	Recover	0 Default: 0 (disabled) Range: 0-86400 seconds

Test the virtual server by generating a significant amount of traffic against the virtual server. Open Attacker (Kali Linux) 100.0.1.3 machine open terminal and type below commands to generate traffic against the virtual server.

curl -s http://100.0.1.100?[1-60]



```
QEMU (Attacker)
root@kali: /home/kali
File Actions Edit View Help
root@kali) - [/home/kali]
# curl -s http://100.0.1.100?[1-60]
```



```
root@kali: /home/kali
File Actions Edit View Help
<h1 style="color:blue;">THIS IS SERVER 3</h1>
</body>
</html>
<html>
<head>
<title>This is Server 3</title>
</head>
<body style="background-color:yellow;">
<h1 style="color:blue;">THIS IS SERVER 3</h1>
</body>
</html>
<html>
<head>
<title>This is Server 3</title>
</head>
<body style="background-color:yellow;">
<h1 style="color:blue;">THIS IS SERVER 3</h1>
</body>
</html>
<html>
<head>
<title>This is Server 3</title>
</head>
<body style="background-color:yellow;">
<h1 style="color:blue;">THIS IS SERVER 3</h1>
</body>
</html>
```

Navigate to Log & Report > Traffic Log > SLB Layer 4&2

Time	Source	Received Bytes	Destination	Sent Bytes	Service	Virtual Server	Duration (s)	Trans Destination	Real Server
15:49:55	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:54	100.0.1.3	395	192.168.2.100	624	tcp	WEB-VS	3	192.168.1.2	SRV2
15:49:53	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:52	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:51	100.0.1.3	395	192.168.2.100	622	tcp	WEB-VS	3	192.168.1.1	SRV1
15:49:51	100.0.1.3	395	192.168.2.100	624	tcp	WEB-VS	3	192.168.1.2	SRV2
15:49:50	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:49	100.0.1.3	395	192.168.2.100	624	tcp	WEB-VS	3	192.168.1.2	SRV2
15:49:49	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:48	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:47	100.0.1.3	395	192.168.2.100	622	tcp	WEB-VS	3	192.168.1.1	SRV1
15:49:47	100.0.1.3	395	192.168.2.100	624	tcp	WEB-VS	3	192.168.1.2	SRV2
15:49:46	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3
15:49:45	100.0.1.3	395	192.168.2.100	624	tcp	WEB-VS	3	192.168.1.2	SRV2
15:49:45	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	3	192.168.1.3	SRV3

Apply the filter SRV3 get more connection as compare to SRV1 and SRV2

Date	Time	Source	Received Bytes	Destination	Sent Bytes	Service	Virtual Server	Trans Destination	Real Server
2024-12-21	15:49:55	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:53	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:52	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:50	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:49	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:48	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:46	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:45	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:44	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:42	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:39	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:38	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:36	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:35	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3
2024-12-21	15:49:34	100.0.1.3	395	192.168.2.100	623	tcp	WEB-VS	192.168.1.3	SRV3

After completing the lab revert back all three servers weight.